

# Near-optimal Linear Predictive Clustering in Non-separable Spaces via MIP and QPBO Reductions

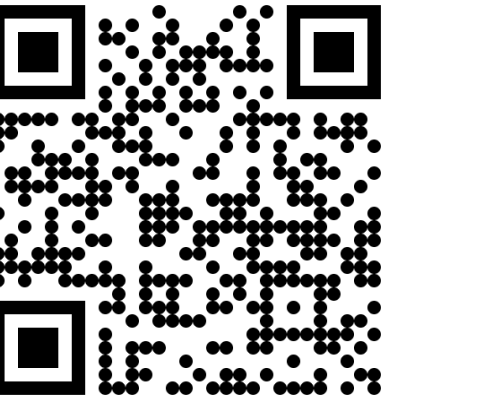
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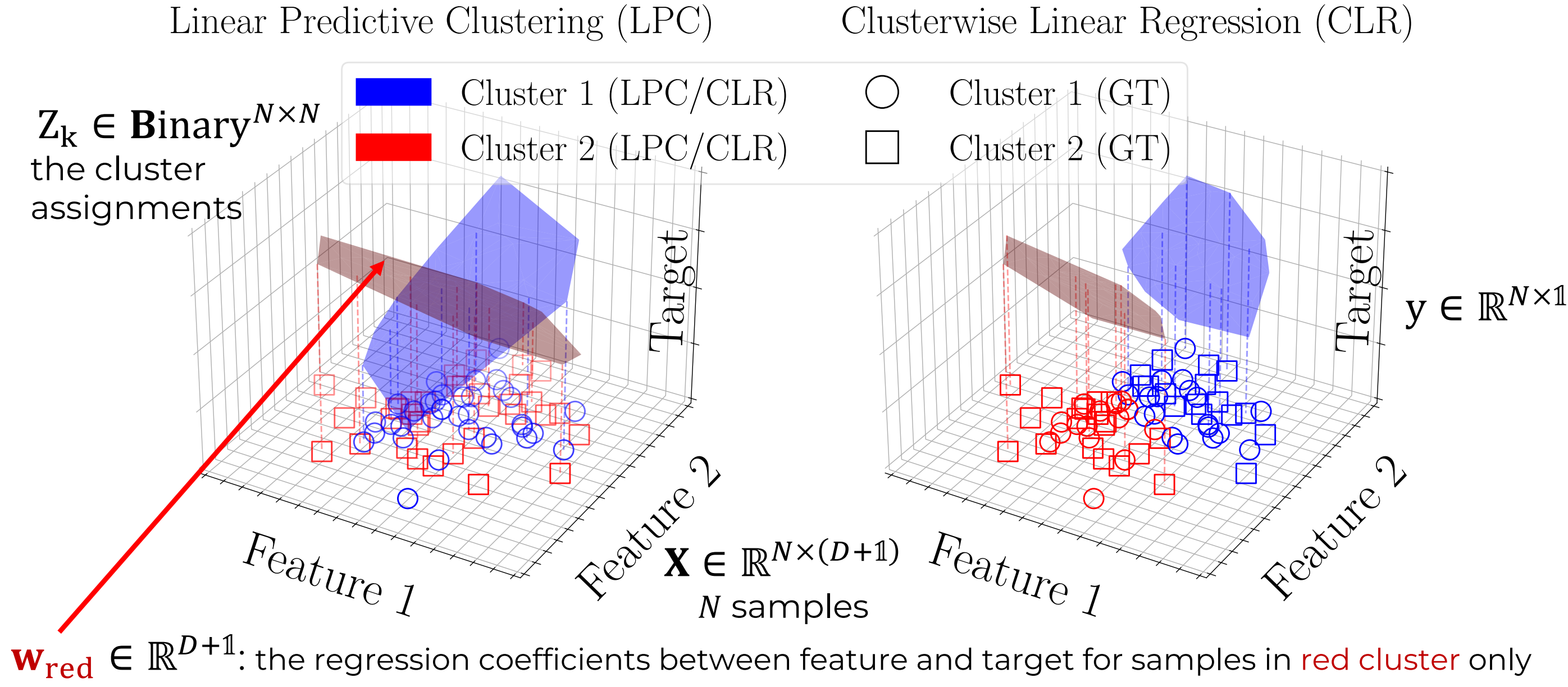


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*TLDR: We solved the most challenging case of Cluster-wise Linear Regression optimally!*

## 1. Introduction

**Linear Predictive Clustering (LPC)** partitions data into  $K$  clusters, where each cluster is defined by a shared linear predictor, enabling the discovery of heterogeneous linear relationships that standard clustering cannot capture.



**Linear Predictive Clustering Objective:**

$$\min_{\mathbf{Z}, \mathbf{w}} \sum_{k=1}^K (|\mathbf{Z}_k \mathbf{y} - \mathbf{Z}_k \mathbf{X} \mathbf{w}_k|_2^2 + \lambda |\mathbf{w}_k|_2^2).$$

LPC extend from the objective of Regularized Linear Least Squares (RLS), determine the cluster assignments and cluster-wise regression coefficients such that SSE across clusters is minimized.

**Non-separable feature space:** different clusters overlap in the feature space.

**Why Cluster-wise Linear Regression is challenge with non-separable feature space:** ignoring differences in the underlying linear relationships with the target variable when assigning cluster.

**We need to optimize  $\mathbf{Z}$  and  $\mathbf{w}$  together!**

(1) **Greedy iterative optimization (Chemhu and Sanner 2023)** which alternates between cluster assignment and confidants: *scales well but is prone to suboptimal noise.*

(2) **Global constrained optimization (Bertsimas and Shioda 2007)** via Mixed-Integer Programming (MIP): *guarantees global optimality but computationally expensive.*

## 2. Methodology

**We all know that there is closed form solution for the Coefficient of RLS:**

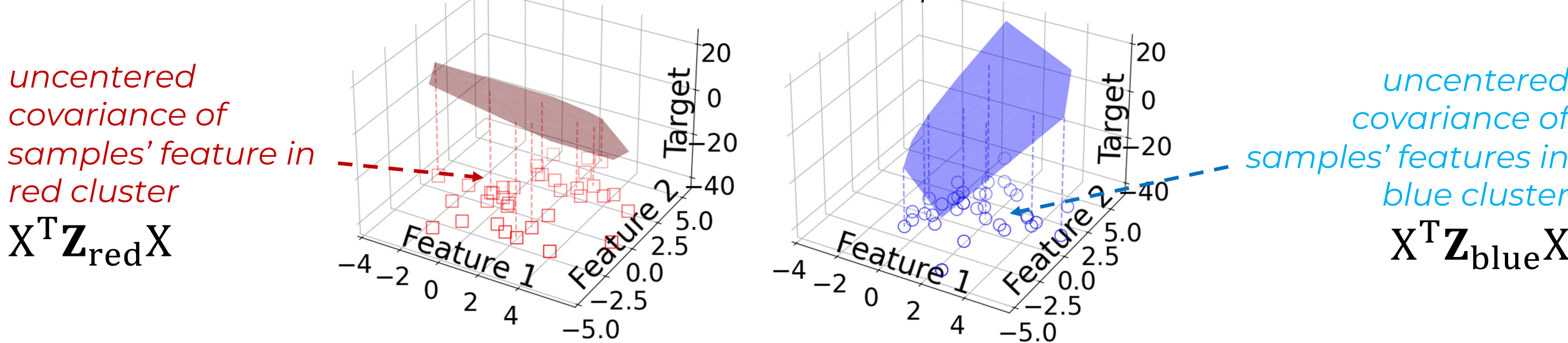
If cluster assignment  $\mathbf{Z}_k$  is given, the there is a closed-form solution for regression coefficients  $\mathbf{w}$ :

$$\mathbf{w}_k = (\underbrace{\mathbf{X}^T \mathbf{Z}_k \mathbf{X} + \lambda \mathbf{I}}_{\mathbf{A}_k})^{-1} \mathbf{X}^T \mathbf{Z}_k \mathbf{y}$$

(a variant for the cluster-wise regression coefficients by introducing  $\mathbf{Z}$ )

**Problem: the inverse of optimization variable  $\mathbf{Z}_k$  is infeasible to formulate as MIP**

In above equation,  $\mathbf{X}^T \mathbf{Z}_k \mathbf{X}$  measures the cluster-wise uncentered covariance of sample in cluster  $k$  in feature variable space



*non-separable infer the uncentered covariance structure across different subsets of samples often remains similar.*

*It may be unnecessary to compute the matrix inversion in  $\mathbf{A}_k$ !*

**Based on this finding, we propose an approximation for  $\mathbf{A}^*$ , defined as:**

$$\mathbf{w}_k^* = (\underbrace{\alpha_k \mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}}_{\mathbf{A}^*})^{-1} \mathbf{X}^T \mathbf{Z}_k \mathbf{y},$$

where  $\alpha_k$  is a hyperparameter reflecting the relative size of cluster  $\mathbf{k}$ .

- Under balanced cluster sizes, this reduces to  $\alpha_k = 1/K$ .
- For imbalanced clusters, a multi-hop tuning procedure yields a more suitable value.

**This leading to a simplified MIP formulation of LPC for non-separate data: LPC-NS-MIP**

We can precomputing  $\mathbf{A}^*$ , reducing the optimization of  $\mathbf{w}$  to a matrix multiplication

$$\min_{\mathbf{Z}} \sum_{k=1}^K (|\mathbf{Z}_k \mathbf{y} - \mathbf{Z}_k \mathbf{X} \mathbf{A}^* \mathbf{X}^T \mathbf{Z}_k \mathbf{y}|_2^2 + \lambda |\mathbf{A}^* \mathbf{X}^T \mathbf{Z}_k \mathbf{y}|_2^2)$$

**LPC-NS-MIP can be further approximated as a Quadratic Pseudo-Boolean Optimization**

We can reformulate and simplify as a maximization problem after expanding the square term:

$$\max_{\mathbf{Z}} \sum_{k=1}^K \left[ \mathbf{y}^T \mathbf{Z}_k \mathbf{X}^T \mathbf{A}^* \left[ \underbrace{2\mathbf{I} - (\mathbf{X}^T \mathbf{Z}_k \mathbf{X} + \lambda \mathbf{I}) \mathbf{A}^*}_{\mathbf{A}_k^{-1}} \right] \mathbf{X}^T \mathbf{Z}_k \mathbf{y} \right]$$

*We can simplify  $[2\mathbf{I} - \mathbf{A}^{*-1} \mathbf{A}^*] = [2\mathbf{I} - \mathbf{I}] = \mathbf{I}$*

yielding the final reduced formulation for **LPC-NS-QPBO**:

$$\begin{aligned} & \max_{\mathbf{Z}} \sum_{k=1}^K [\mathbf{y}^T \mathbf{Z}_k \mathbf{X}^T \mathbf{A}^* \mathbf{X}^T \mathbf{Z}_k \mathbf{y}] \\ & \text{s. t. } \sum_{k=1}^K \mathbf{Z}_{k,i,i} = 1, \mathbf{Z}_{k,i,i} \in \{0,1\}; \\ & \quad \forall i \in \{1, \dots, N\}; k \in \{1, \dots, K\} \end{aligned}$$

Note:  $\mathbf{X}^T \mathbf{A}^* \mathbf{X}^T$  does not involves about optimization variables, can be precomputed

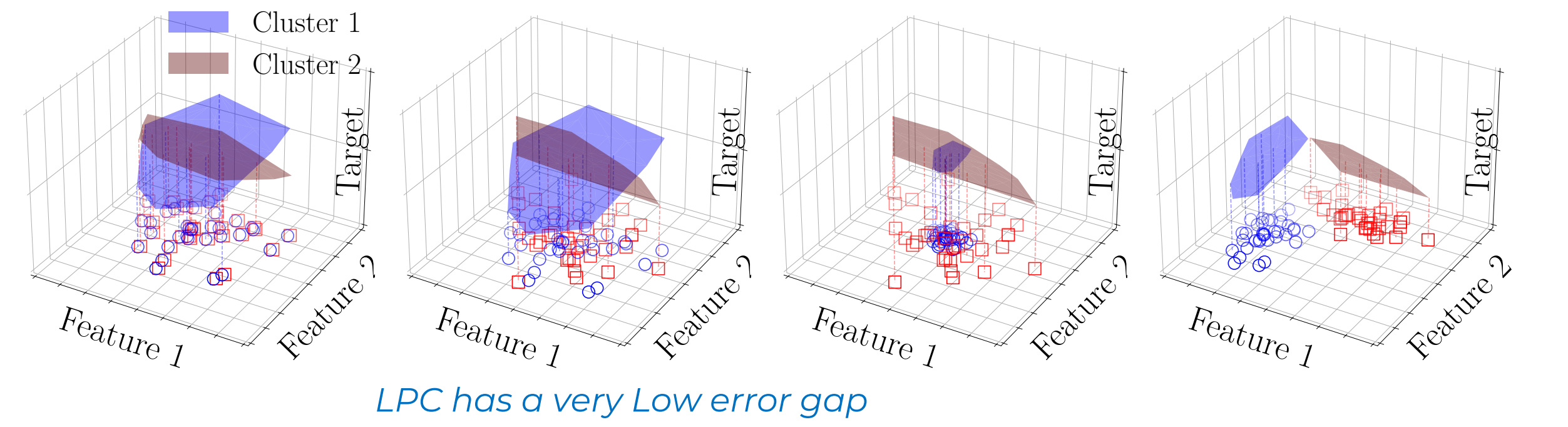
## 3. Approximation Error of $\mathbf{w}^*$

it is crucial to both theoretically and empirically assess the bound of the approximation error in the LPC objective introduced by approximating  $\mathbf{w}$  as  $\mathbf{w}^*$

We found the upper bound of the **difference** between the **globally optimal objective** and the **objective using  $\mathbf{w}^*$**  is

$$\epsilon_{\text{obj}} \leq \sum_{k=1}^K \left( \frac{1}{2\lambda^2} 3|\mathbf{y}_k|_2^2 |\mathbf{X}_k|_2^2 \underbrace{|\alpha_k \mathbf{X}^T \mathbf{X} - \mathbf{X}^T \mathbf{Z}_k \mathbf{X}|_2}_{\epsilon_{\text{sep}} = 0} \right)$$

when the approximate uncenter covariance and actual covariances are the same (clusters are identical in feature space)



|                         | Identical           | Non-Separable                         | Semi-Separable       | Separable           |
|-------------------------|---------------------|---------------------------------------|----------------------|---------------------|
| $\epsilon_{\text{sep}}$ | 0                   | 69.404                                | 193.791              | 666.579             |
| $\epsilon_{\text{obj}}$ | $0.01 \pm 0.01$     | <b><math>0.0208 \pm 0.0054</math></b> | $2.7080 \pm 0.04973$ | $2.6250 \pm 0.4868$ |
| $\epsilon_{\text{obj}}$ | $18.4862 \pm 0.441$ | $72.2481 \pm 0.8360$                  | $9.4433 \pm 0.3633$  | $0.5996 \pm 0.0316$ |

*Existing Method did well on separable cases! But failed on more challenges cases.*

## 4. Experiments and Results

**We measures following metrics to compare the performance across methods:**

**Difference From Optimal Objective**

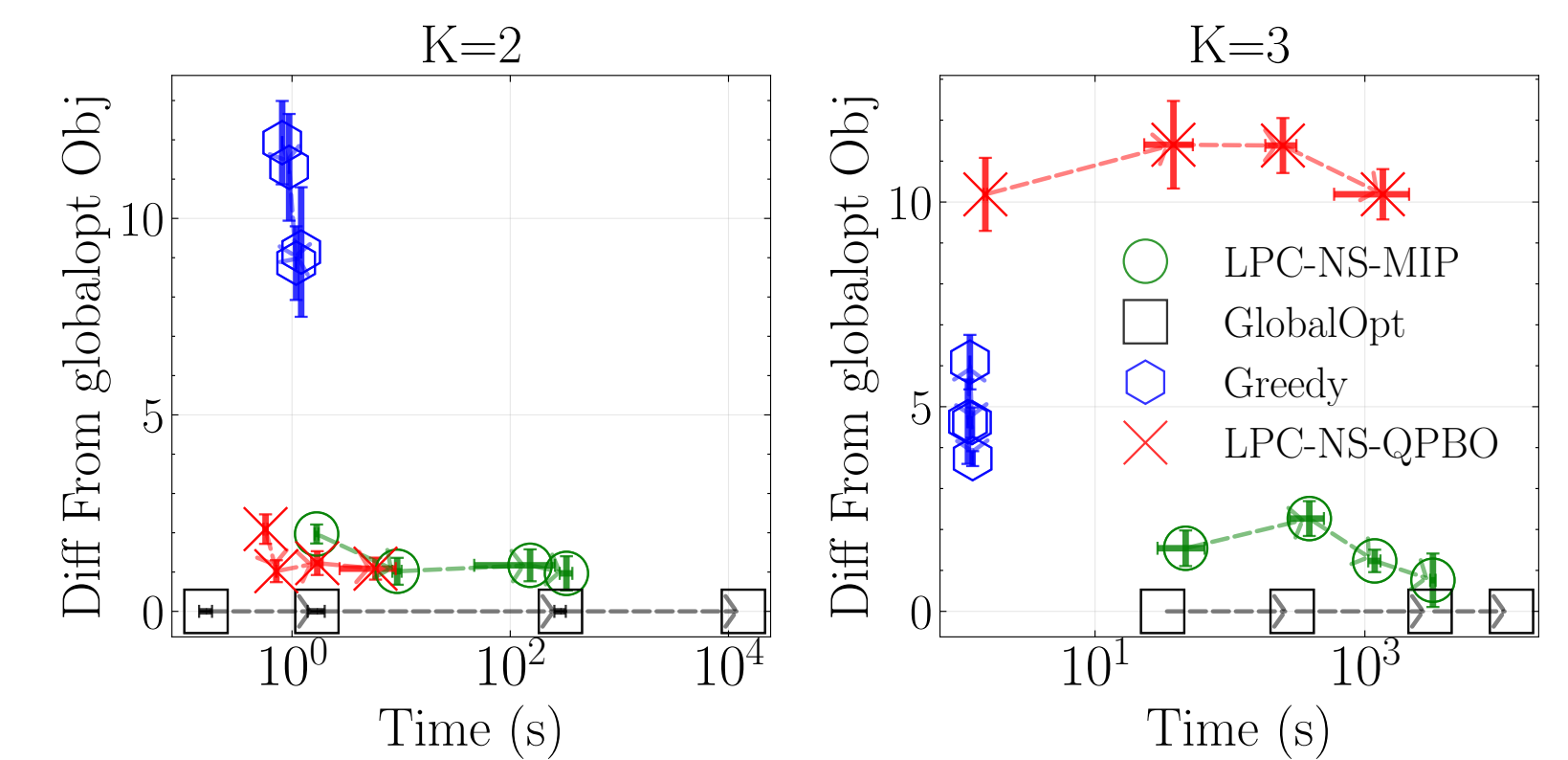
Difference between the the global optimal LPC objective

**Cluster Label Difference**  $\frac{1}{N} \sum_{k=1}^K |\mathbf{Z}_k^{\text{gt}} - \mathbf{Z}_k'|_1$   
Fraction of misassigned samples.

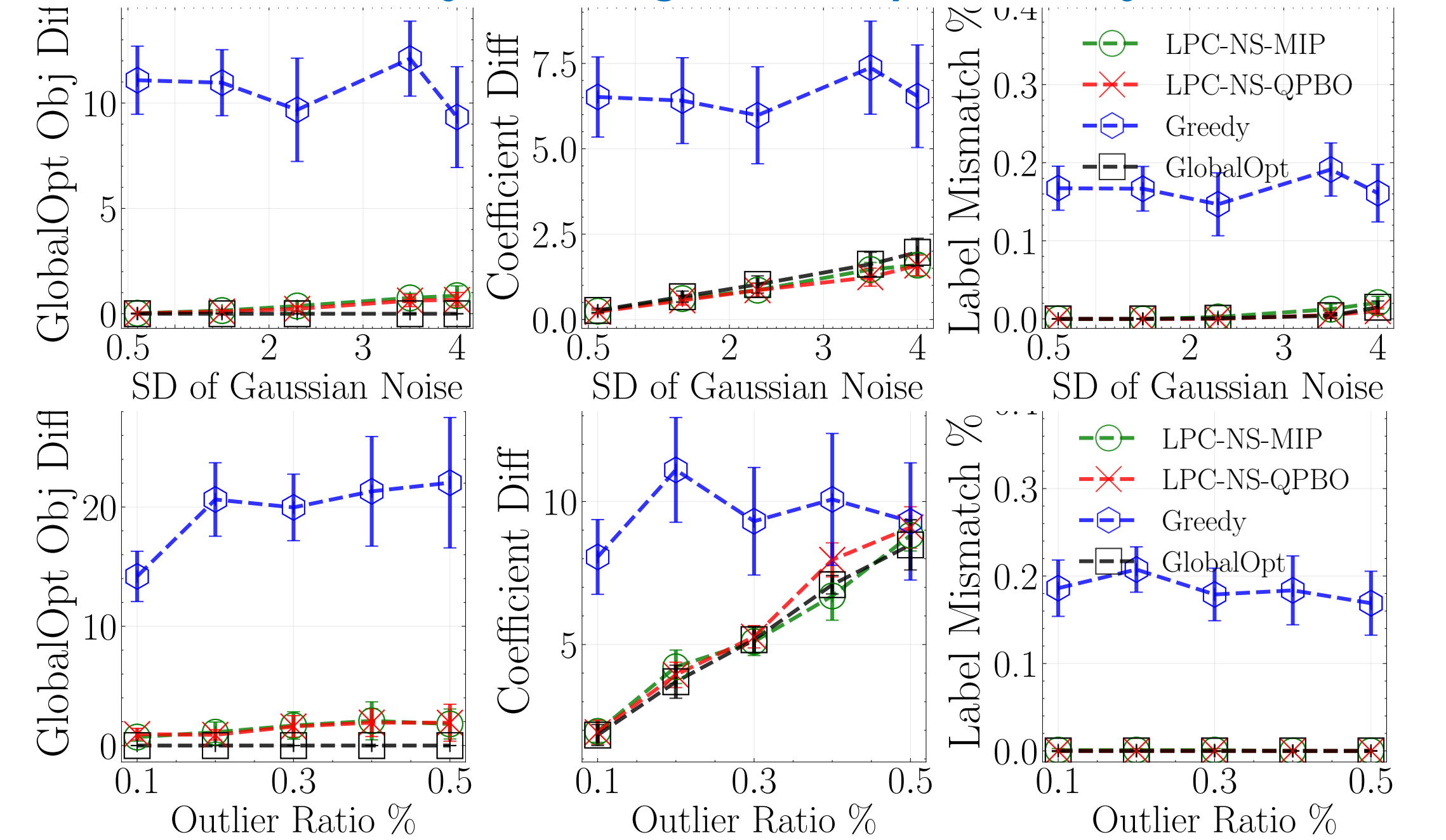
**Regression Coefficient Difference**  $\sum_{k=1}^K |\mathbf{w}_k^{\text{gt}} - \mathbf{w}_k'|_2$

Error in recovered the ground truth cluster wise linear models

**RQ1: Time vs. Error Scalability Trade-off Analysis**



**RQ2: Performance Analysis and Algorithm Properties on Synthetic Datasets.**



**RQ3: Performance Analysis on Real Datasets.** How do different methods compare in terms of objective error on real datasets?

| Dataset (UCI ID)               | LPC-NS-QPBO     | LPC-NS-MIP      | Greedy                | Separability |
|--------------------------------|-----------------|-----------------|-----------------------|--------------|
| Stock Performance (390)        | 16.1798         | <b>12.4745</b>  | $17.1278 \pm 1.6503$  | 0.0000       |
| Servo (87)                     | 12.4865         | <b>2.5305</b>   | $4.0431 \pm 1.1435$   | 0.0543       |
| Solar Flare (89)               | <b>3.8289</b>   | 11.0511         | $8.3014 \pm 1.0420$   | 0.5885       |
| Productivity Prediction (597)  | 41.4977         | <b>38.2234</b>  | $43.4662 \pm 2.7841$  | 0.5914       |
| Heart Failure Prediction (519) | 4.73E-30        | <b>6.92E-32</b> | $25.0644 \pm 8.6361$  | 0.7685       |
| Liver Disorders (60)           | 108.3983        | <b>94.9041</b>  | $112.0968 \pm 5.4914$ | 0.9784       |
| Student Performance (320)      | <b>4.8812</b>   | 5.6557          | $6.8896 \pm 0.4574$   | 2.0107       |
| Park Telemonitoring (189)      | <b>135.3372</b> | 146.9704        | $174.642 \pm 12.7622$ | 3.9580       |
| Facebook Metrics (368)         | 2.88E-20        | <b>2.05E-20</b> | $3.1E-5 \pm 1.2E-5$   | 5.6163       |
| Infrared Thermography (925)    | 1.08E-13        | <b>6.90E-15</b> | $0.1029 \pm 0.0380$   | 7.9983       |

## Conclusion

**LPC-NS-MIP:** efficient than GlobalOpt, yet near-optimal in non-separable  
**LPC-NS-QPBO:** improves scalability via QPBO reduction

Across noise/dimension/outlier sweeps, LPC-NS-MIP > Greedy for objective optimization and for recovering cluster assignments + regression coefficient  
LPC-NS-QPBO remains highly competitive for K=2 and significantly scalable than LPC-NS-MIP